

Umesh Kumar, PhD

Contact Details

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Nationality: Indian

Profile Links: [Google Scholar](#); [LinkedIn](#); [Orchid](#); [PubMed](#); [Scopus](#); [ResearchGate](#); [Web of Science](#).

Address: Lucknow, India

D.O.B: 01st February 1991

Skill: Metabolomics, Biomarker Discovery, Python/R

Professional Profile

Metabolomics researcher with experience in NMR, LC-MS/MS, and GC-MS/MS-based metabolomics and Lipidomics. Skilled in method development, large-scale clinical and pre-clinical sample analysis, and multivariate statistical analysis. Proven publication record (30+ papers, h-index 21, citations ~1200) and international postdoctoral experience (Austria). Passionate about biomarker discovery, translational metabolomics, and precision medicine. Currently seeking a Postdoctoral/Scientist position to advance metabolomics-driven biomedical research.

Work Experience

Postdoc Fellow & Guest Lecturer

SANJAY GANDHI POSTGRADUATE INSTITUTE OF MEDICAL SCIENCES

Lucknow, India | September 2024 – May 2026

- Delivered teaching and practical training in Analytical Research Methods including LC-MS/GC-MS, NMR, Biochemistry, Metabolism, and Multivariate Statistics using Python/R.
- Conducted research in metabolomics and clinical science focusing on pancreatic cancer and AKI/CKD for biomarker and pathway analysis.
- Developed LCMS based SOPs for biofluid metabolomics, Lipidomics and contributed to collaborative data analysis, interpretation, and publications.

Postdoctoral Researcher

UNIVERSITY OF INNSBRUCK

Innsbruck, Austria | May 2023 – June 2024

- Developed and optimized SOPs for stool, saliva, and urine metabolomics (NMR-based).
- Processed and analysed >1200 clinical samples under the VAScage programme on vascular health.
- Collaborated with clinicians and interdisciplinary researchers, contributing to high-quality manuscripts.

Research Associate-III

NATIONAL BUREAU OF ANIMAL GENETIC RESOURCES

Karnal, India | Dec 2021 – March 2023

- Characterized Ladakhi cow and yak colostrum using NMR and LC/GC-MS-based metabolomics.
- Conducted RNA isolation, antioxidant assays, ELISA, flow cytometry, microscopy, and animal handling.
- Supported a national-level mega project on identifying biomolecules with therapeutic potential.

Senior Research Fellow (PhD)

CENTRE OF BIOMEDICAL RESEARCH (CBMR)

Lucknow, India | September 2015 – October 2021

- Identified metabolic signatures of autoimmune inflammatory diseases (Takayasu arteritis, Lupus

Nephritis).

- Carried out NMR spectroscopy (400–800 MHz) and multivariate statistical analyses.
- Drafted manuscripts, guided MSc dissertations, and collaborated with clinical partners.

Project Assistant- Parasitology Lab

BAREILLY COLLEGE BAREILLY (MJPR UNIVERSITY)

Bareilly, India | May 2014 – August 2015

- Specialized in collection and isolation of root-knot nematode, *Meloidogyne incognita*.
- Prepared samples for electron microscopy and SEM analysis.
- Investigated nematocidal effects of herbal extract.

Summer Intern

Indian Institute of Science (IISc)

Bangalore, India | May 2011 – July 2011

- **Project:** *Molecular Phylogeny of Hemidactylus brookii species complex in the Himalayas region.*
- Performed DNA isolation and purification of *Hemidactylus brookii* as Summer Intern.
- Supported research team with electrophoresis procedures.
- Analysed genetic data for molecular phylogeny during summer internship

Education

- **PhD (Clinical Metabolomics)** – Centre of Biomedical Research, Lucknow, India (2015–2020)
- **MSc (Biological Science)** – BBA University (Central University), Lucknow, India (2010–2012)
- **BSc (Biology/Chemistry)** – MJP Rohilkhand University, Bareilly, India (2007–2010)

Technical Key Skills

NMR based Metabolomics

- **Analytical Skills-** 1D & 2D-NMR data acquisition from Solution-state 400, 600 & 800 MHz NMR ¹H-NMR (CPMG, ZGPR, NOESY etc), ¹H-¹H NMR (JRES, COSY), ¹H-¹³C NMR (HSQC, HMBC and TOCSY).
- **Computational Skills** - Topspin, Amix Aurelia (spectral bucketing), Chenomx (Conc. profiling).

LCMS/GCMS Based Metabolomics and Lipidomics

- **Analytical Skills** -HPLC (Agilent &Water), LC-MS/MS (Agilent &Water) and GC-MS/MS (Shimadzu)
- **Computational Skills**- MetaboAnalyst, Mass lynx, Mass hunter, MS-Omics, MS-Dial, XCMS, lipid maps.

Multivariate Statistical Analysis (MVSA)

- **Data Analysis and Visualization:** Python/R (basic), MetaboAnalyst, SIMCA, GraphPad Prism, Excel
- **Other tools:** SIGMA plot, Venny 2.1.0, Chiplot, Biorender, Chem Biodraw, Adobe Illustrator, Origin etc.
- **Database:** HMDB, Reactome Pathway Database, Pathbank, Pathviews

Other Tools/Techniques

- **Instrument:** FTIR, UV, Microscopes, Centrifugation machines, Lyophilizer, Sonication etc.
- **Biological Skills:** Preparation of serum, plasma, urine, faeces, saliva, milk, tissue biopsy; cell culture; animal handling
- **Molecular Biology:** ELISA, Antioxidant Assay, DNA/RNA Isolation.

Trainings/ Additional Roles:

Analytical Trainee – Fare Labs Pvt Ltd, India (April-July 2025)

- Handling CRM, stock management, and QA/QC support for laboratory operations.
- Performed pesticide/antibiotic residue detection in cereals, milk, and vegetables using LC-MS/MS and GC-MS/MS.
- Supported method development, instrument calibration, and troubleshooting.

Awards & Recognitions

- **Indian Council of Medical Research-Junior Research Fellow- 2015**
- **Indian Council of Medical Research -Senior Research Fellow-2018**
- **CSIR NET (Lectureship, 2015)**
- **Review Editor – *Frontiers in Molecular Biosciences***

Key Publications

- SK Shekhar, **Umesh Kumar**. *et al.* ^1H NMR-based serum metabolomics identifies discriminatory metabolic signatures associated with pancreatic cancer. *Scientific Reports* **16**, 17919 (2026). [Link](#)
- Herzog, Chiara, Charlotte DV, Elisa Redl, Hagen M, Gabriel Knoll, Christina Watschinger, **Umesh Kumar** et al. **Multi-modal atlas of lifestyle interventions reveals malleability of ageing-linked molecular features.** *bioRxiv* (2025): 2025-08.) [Link](#)
- Herzog, Chiara Maria Stella, Charlotte D. Vavourakis, Bente Theeuwes, Elisa Redl, Christina Watschinger, Gabriel Knoll, **Kumar U**, Magdalena Hagen et al. **Systemic multi-omic remodelling underlies health benefits of intermittent fasting.** *bioRxiv* (2025). [Link](#)
- Herzog CM, Vavourakis CD, Theeuwes B, Redl E, Watschinger C, Knoll G, Hagen M, Haider A, Platzer HP, **Kumar U**, Zollner-Kiechl S. **Longitudinal multi-omic evaluation of biomarkers of health and ageing over smoking cessation intervention.** *bioRxiv*. 2025 Sep 4:2025-08. [Link](#)
- Garg, V., Mukesh, M., **Kumar, U.**, Kumar, D., ... & Sodhi, M. (2025). **Characterization of metabolite profiles in milk derived exosomes from indicus, crossbred and taurine cows by proton nuclear magnetic resonance analysis.** *Food Chemistry* (**IF 10.3**). [Link](#)
- **Kumar U. et al.** (2024) **Characterizing metabolome signature of colostrum, transition and mature milk of indigenous cows at high altitude.** *Food Chemistry* (**IF 10.3**). [Link](#)
- Shekher A, Puneet, Awasthee N, **Kumar U**, Raj R, Kumar D, Gupta SC (2023). **Association of altered metabolic profiles and long non-coding RNAs expression with disease severity in breast cancer patients: analysis by ^1H NMR spectroscopy and RT-q-PCR.** *Metabolomics*. (**IF 3.9**). [Link](#)
- Arya, P., **Kumar, U.**,... & Kumar, D. (2021). **Targeted NMR-based serum metabolic profiling of serine, glycine and methionine in acute-on-chronic liver failure patients: possible insights into mitochondrial dysfunction.** *Analytical Science Advances*. (**IF 5.4**). [Link](#)
- Guleria, A., **Kumar, U.**, Kumar, D.,... & Gupta, L. (2021). **NMR-based serum and muscle metabolomics for diagnosis and activity assessment in idiopathic inflammatory myopathies.** *Analytical Science Advances*. (**IF 5.4**). [Link](#)
- Nisha, R., Kumar, P., **Kumar, U.**,... & Saraf, S. A. (2022). **Assessment of hyaluronic acid-modified imatinib mesylate cubosomes through CD44 targeted drug delivery in NDEA-induced hepatic carcinoma.** *International Journal of Pharmaceutics*. (**IF 6.0**). [Link](#)

- Kumar U. *et al.* (2021) **Metabolic derangements in patients with mild cognitive impairment.** *Metabolic Brain Disease* (IF 3.9). [Link](#)
- Kumar U. *et al.* (2021) **Histidine levels as predictive indicators of disease activity in Takayasu arteritis.** *Analytical Science Advances*. (IF 5.4). [Link](#)
- Kumar U. *et al.* (2020) **Glutamine/glucose ratio for evaluating disease activity in Takayasu arteritis.** *Journal of Pharmaceutical and Biomedical Analysis* (IF 3.4). [Link](#)
- Kumar, U. *et al.* (2021) **Serum metabolic disturbances associated with acute-on-chronic liver failure in patients with underlying alcoholic liver diseases: An elaborative NMR-based metabolomics study.** *Journal of Pharmacy and Bioallied Sciences*. [Link](#)
- Guleria A., Kumar U. *et al.* (2016) **NMR-based serum metabolomics reveals a distinctive signature in Lupus Nephritis.** *Scientific Reports* (IF 3.9). [Link](#)

➤ **Summary of Publications (Journal, IF, Frequency)**

Journal Name	IF	Frq.	Journal Name	IF	Frq.
Scientific Report	4.3	4	Food Chemistry (ACS)	10.4	2
Journal of Proteomics Research	3.8	3	Annals of the Rheumatic Diseases	24.0	3
Analytical Science Advances	5.4	3	Arthritis & Rheumatology	9.9	3
Metabolomics	3.9	1	Pharmacological Research	12.2	1
Molecular Pharmaceutics	4.9	1	Life Sciences	6.4	1
International Journal of Pharmaceutics	6.0	1	Biomedicine & pharmacotherapy	7.5	1

➤ Full list of publications: [Google Scholar](#)

Research Vision

"My long-term goal is to lead innovative metabolomics research that uncovers disease mechanisms, identifies clinically actionable biomarkers, and accelerates the translation of omics science into personalized healthcare solutions".

~Umesh

Reference/Recommendations

Available upon request

Place, Date: Lucknow, June 25, 2026